

Lab Exercise 19- ClusterIP in Service in Kubernetes

A ClusterIP is the default type of service in Kubernetes. It provides a stable internal IP address for the service, which is accessible only within the cluster. This is useful for internal communication between services within the cluster.

Step 1: Create Deployment (Correct Labels)

Create file:

nginx-deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx # MUST match template labels
  template:
    metadata:
      labels:
        app: nginx # MUST match service selector
    spec:
      containers:
        - name: nginx
          image: nginx
          ports:
            - containerPort: 80
```

Save file.

```
kubectl apply -f nginx-deployment.yaml
```

```
devanksilswal@devanks-MacBook-Air ex_s % touch nginx-deployment.yaml
devanksilswal@devanks-MacBook-Air ex_s % nano nginx-deployment.yaml
devanksilswal@devanks-MacBook-Air ex_s % kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-app created
```

Verify:

```
kubectl get pods --show-labels
```

```
devanksilswal@devanks-MacBook-Air ex_s % kubectl get pods --show-labels
NAME                                READY   STATUS    RESTARTS   AGE   LABELS
nginx-app-7cccd94f7-qnw22          1/1     Running   0           2m41s   app=nginx,pod-template-hash=7cccd94f7
nginx-app-7cccd94f7-wr7v7          1/1     Running   0           2m41s   app=nginx,pod-template-hash=7cccd94f7
```

You should see:

```
app=nginx
```

Step 2: Create ClusterIP Service (Matching Selector)

Create file:

nginx-service.yaml

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  type: ClusterIP
  selector:
    app: nginx # MUST match pod label
  ports:
```

```
- port: 80
targetPort: 80
```

Apply:

```
kubectl apply -f nginx-service.yaml
```

```
[devansilswal@devanks-MacBook-Air ex_s % touch nginx-service.yaml
[devansilswal@devanks-MacBook-Air ex_s % nano nginx-service.yaml
[devansilswal@devanks-MacBook-Air ex_s % kubectl apply -f nginx-service.yaml
service/nginx-service created
```

Step 3: Verify It Is Working

Check Service

```
kubectl get svc
```

```
[devansilswal@devanks-MacBook-Air ex_s % kubectl get svc
NAME                TYPE          CLUSTER-IP    EXTERNAL-IP  PORT(S)    AGE
nginx-service       ClusterIP     10.109.159.166 <none>       80/TCP     5s
```

You should see:

```
TYPE          CLUSTER-IP
ClusterIP     10.x.x.x
```

MOST IMPORTANT CHECK

```
kubectl get endpoints nginx-service
```

```
[devansilswal@devanks-MacBook-Air ex_s % kubectl get endpoints nginx-service
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME                ENDPOINTS                                AGE
nginx-service       10.1.0.88:80,10.1.0.89:80              14s
```

Now you should see something like:

10.244.x.x:80, 10.244.x.x:80

Step 4: Test From Inside Cluster

Run test pod:

```
kubectl run test-pod --image=busybox --rm -it -- sh
```

```
[devanksilswal@devanks-MacBook-Air ex_s % kubectl run test-pod --image=busybox --rm -it -- sh
All commands and output from this session will be recorded in container logs, including credentials and sensitive information passed through the command prompt.
If you don't see a command prompt, try pressing enter.
[/ # wget -qO- http://nginx-service
```

Inside pod:

```
wget -qO- http://nginx-service
```

You should see:

```
Welcome to nginx!
```

Exit:

```
exit
```

```
[/ # wget -qO- http://nginx-service
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
[/ # exit
```

Step 5: Test From Outside (It Will Fail)

From your local machine:

```
curl http://<CLUSTER-IP>
```

It will NOT work.

That proves:

ClusterIP works only inside cluster.

```
[devanksilswal@devanks-MacBook-Air ex_s % curl http://10.109.159.166
curl: (28) Failed to connect to 10.109.159.166 port 80 after 75004 ms: Couldn't connect to server
```